Ticketing Platform

Features List and Core Flow

Version 1.0 (Draft) | Prepared for the product and development team

Purpose. This document lists what the platform will do, who does it, in what order, and which release each feature belongs to. **MVP** = must exist for the first working version. **V2** = planned right after the core flow works. **Later** = nice-to-have.

1. Product in one paragraph

A web platform where event organizers create events, an admin approves them, and the public buys tickets through a shareable link (as a guest or a logged-in user). Each ticket is delivered as a PDF with a unique QR code and is verified once at the entrance by volunteers. Organizers get live sales dashboards and an end-of-event report. Payment uses a dummy bKash/Nagad-style gateway with OTP in this version.

2. Users and roles

Role	Who	What they can do
Admin	Platform staff	Review and approve/reject events, view all events, receive notifications, suspend events or organizers.
Organizer (Owner)	Person who creates the event	Create/edit events, define ticket categories and form, invite team, view sales and reports, close the event.
Manager	Teammate invited by Owner	Edit event details, view sales, dashboard and reports, manage attendee list.
Scanner (Volunteer)	Teammate invited by Owner	Check-in only: scan QR or type ticket ID, see valid/invalid/already used.
Viewer	Teammate invited by Owner	Read-only access to dashboard and reports.
Buyer (Registered)	Public user with account	Buy tickets with prefilled email, see order history, re-download tickets.
Buyer (Guest)	Public user without account	Buy tickets with email only, find tickets later via email + OTP.

3. Core flow (end to end)

The main journey has eight stages. Every other feature supports one of these stages.

#	Stage	What happens	Actor
1	Organizer sign-up	Organizer registers/logs in with email and password (Google login optional later) and verifies email.	Organizer
2	Create event	Organizer fills the event form: name, description, venue, banner, date and time, sales start/end, ticket categories with price and seat limit, max tickets per person, optional custom fields. Saved as Draft, then Submitted.	Organizer
3	Admin approval	Admin gets a notification, reviews the event, and approves or rejects with a reason. Organizer is notified by email either way. On approval, a public event link is generated.	Admin
4	Public event page	Anyone opens the link. Page shows banner, details, categories, prices, and remaining availability. Link previews well on Facebook/WhatsApp.	Buyer
5	Buy tickets	Buyer chooses guest or login, selects category and quantity, fills email and (if enabled) Attendee 1, Attendee 2... names and other fields. Seats are held for 10 minutes. Buyer pays via dummy gateway and enters OTP.	Buyer
6	Ticket delivery	On confirmed payment the system creates tickets with unique QR codes and one PDF per order (one page per ticket). PDF is emailed to the recipient and downloadable on the confirmation page. Buyer sees a confirmation screen and receipt.	System
7	Entry check-in	At the gate, volunteers scan the QR or type the ticket ID. First valid scan marks the ticket as used; any repeat shows 'Already used' with time and scanner name.	Scanner
8	Close and report	After the event, organizer closes it and receives a report: sold vs unsold by category, revenue, attendance (checked in vs no-show).	Organizer

Event status lifecycle

Status	Meaning	Next step
Draft	Organizer is still editing; not visible to admin or public.	Submit for approval
Pending approval	Waiting for admin review.	Approved or Rejected
Rejected	Admin gave a reason; organizer can edit and resubmit.	Back to Draft
Approved	Approved; link exists. Sales open at the sales start time.	On sale
On sale	Tickets purchasable between sales start and end.	Sales closed
Sales closed	No more purchases; check-in still works.	Event ended
Event ended	Organizer closes the event; report is generated.	Report ready

Ticket status lifecycle

Status	Meaning
Held	Seat reserved for a buyer for 10 minutes while paying. Released automatically if payment does not finish.
Valid	Payment confirmed; ticket issued and can be used.
Used	Checked in at entry. Cannot be used again.
Cancelled	Cancelled by the organizer (refunds are handled outside the platform in this version). Scanning shows 'Cancelled'.

4. Feature list by module

Priority: **MVP** (first release), **V2** (next), **Later** (future).

4.1 Accounts and authentication

Feature	Details	Priority
Organizer sign-up/login	Email + password, email verification, password reset.	MVP
Buyer login (optional)	Login with email; email field prefilled on checkout.	MVP
Guest checkout	Buy with only an email; no account needed.	MVP
Google login	One-click sign-in for organizers and buyers.	V2
Profile	Name, phone, change password.	MVP

4.2 Event creation (form builder)

Feature	Details	Priority
Core fields	Name, description, venue, banner image, date and time (shown in Asia/Dhaka).	MVP
Sales window	Separate sales start and end time from the event time.	MVP
Ticket categories	Organizer adds categories (Front, Regular, Student, Seating, Standing...) each with price and its own seat limit (default unlimited).	MVP
Per-person limit	Max tickets one buyer can buy per order.	MVP
Attendee name toggle	If on, checkout shows Attendee 1 name, Attendee 2 name... per ticket.	MVP
Custom fields	Organizer adds optional boxes (text, number, dropdown, checkbox), marked required or optional.	MVP
Simple events	Events with no extra fields: just quantity, email, payment, done.	MVP
Save draft / preview	Save and preview the public page before submitting.	MVP
Edit after approval	Allowed with rules; changing date or venue notifies ticket holders.	V2
Duplicate event	Copy a previous event as a starting point.	Later

4.3 Admin approval

Feature	Details	Priority
Approval queue	List of pending events with full details.	MVP
Approve / reject	Reject requires a reason that is emailed to the organizer.	MVP
Notifications	In-app and email notification when an event is submitted.	MVP
Link generation	Unique, readable public link after approval.	MVP
Suspend event/organizer	Stop sales on abuse or fraud.	V2
Audit log	Record of every admin action.	V2

4.4 Public event page and purchase

Feature	Details	Priority
Event page	Banner, details, categories, prices, live availability, sold-out labels. Server-rendered with share preview.	MVP
Guest or login choice	Two clear options at checkout.	MVP
Ticket selection	Choose categories and quantities within limits.	MVP
Seat hold	Selected seats reserved for 10 minutes; no overselling.	MVP
Gifting	Buyer can enter another person's email as the recipient; recipient gets the PDF, buyer gets a confirmation screen and receipt.	MVP
Order summary	Review before paying.	MVP
Promo codes	Discount codes per event.	Later
Waitlist	Join a waitlist when sold out.	Later

4.5 Payment (dummy gateway)

Feature	Details	Priority
Provider choice	bKash / Nagad / other mobile banking option (simulated).	MVP
Phone number + OTP	Simulated OTP step; success confirms the order.	MVP
Failure and timeout	Failed or abandoned payment releases held seats.	MVP
Payment records	Every attempt stored with status and reference.	MVP
Swappable gateway	Payment module built behind a clean interface for a real provider later.	MVP
Real gateway	bKash/Nagad API or aggregator integration.	Later

4.6 Tickets, PDF and email

Feature	Details	Priority
Unique ticket ID	Human-friendly ID plus a secure random token.	MVP
Unique QR code	Encodes the secure token (not a guessable number); stored with ticket ID and details.	MVP
Ticket PDF	Fixed branded design; one PDF per order with one page per ticket. Shows event, date/time, venue, category, attendee name, ticket ID, QR.	MVP
Email delivery	Sent from the platform's email address to the recipient; send runs in the background.	MVP
Download	Ticket PDF downloadable from the confirmation page.	MVP
Find my ticket	Guests retrieve tickets with email + emailed OTP; resend email option.	MVP
My orders	Logged-in buyers see order history.	V2

4.7 Check-in and entry

Feature	Details	Priority
QR scan	Mobile web page uses phone camera to scan.	MVP
Manual entry	Type the ticket ID if QR cannot be read.	MVP
Single-use rule	Atomic check: first scan wins; repeat shows used time and scanner name.	MVP
Clear result screen	Green (valid) or red (invalid/used/cancelled) with name and category.	MVP
Live attendance count	Checked-in vs total, updating live.	MVP
Offline fallback	Downloadable attendee list for poor venue internet.	V2

4.8 Team and roles

Feature	Details	Priority
Invite by email	Owner enters teammate emails; they receive an invitation link and accept it.	MVP
Role assignment	Manager, Scanner, Viewer (per event).	MVP
Revoke access	Owner can remove a member any time.	MVP
Activity trail	See who scanned which ticket and when.	V2

4.9 Dashboard and reports

Feature	Details	Priority
Basic dashboard charts	Tickets sold over time, sales by category, revenue, check-in progress.	MVP
Final report	Total sold, unsold, by category, revenue, checked-in vs no-show.	MVP
Export	CSV export of attendees and orders.	V2
Report download	Report as PDF.	V2

4.10 Platform and security

Feature	Details	Priority
Time zones	Stored in UTC, displayed in Asia/Dhaka.	MVP
Rate limiting	Limits on OTP, login and purchase endpoints.	MVP
Image validation	Banner type and size checks.	MVP
Overselling protection	Database transactions and locking on seat counts.	MVP
Privacy	Buyer data visible only to that event's organizer team.	MVP

5. Out of scope for this version

- Refund and cancellation handling inside the platform (organizer and attendee settle directly).
- Payouts, platform fees and commission tracking (dummy payments are treated as going to the organizer).

- Real payment gateway integration.
- Native mobile apps (the scanner is a mobile-friendly web page).
- Reserved seat maps (seating is handled as category limits only).

6. Suggested build phases

Phase	Focus	Outcome
1	Auth, event form, categories, admin approval, public page	Event can be created, approved and viewed
2	Seat hold, order, dummy payment, ticket + QR, PDF, email	A buyer can purchase and receive a ticket
3	Check-in scanner, single-use rule, team invites and roles	Event can run on the day
4	Dashboard, final report, find-my-ticket, polish	Organizer gets insights after the event
5	V2 items (Google login, CSV, offline list, edit rules)	Hardening and extras